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Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Neurofibrillary tangles seen in Alzheimer's disease are made up of

Select one:

- Hypophosphorylated tau protein
- Amyloid fibrils
- Actin filaments
- Synaptophysin
- Hyperphosphorylated tau protein

Neurofibrillary tangles are composed of cytoskeletal elements, primarily abnormally hyperphosphorylated tau protein. Tau is a peptide required for microtubule assembly. Microtubules are essential to transport of materials down axons. Hyperphosphorylation of Tau proteins can result in the self-assembly of tangles of paired helical filaments and straight filaments, which are involved in the pathogenesis of Alzheimer's disease.

The correct answer is: Hyperphosphorylated tau protein

Question **2**

Not answered

Marked out of 1.00

Flag question

The pathological factor that correlates most with cognitive decline in Alzheimer's disease is

Select one:

- Burden of diffuse plaques
- Burden of neurofibrillary tangles
- Burden of neuritic plaques
- Burden of gliosis
- Burden of vascular amyloid load

A staging scheme devised by Braak and Braak (1995) is widely used to describe the extent of tangle related abnormalities (distribution from entorhinal cortex to isocortex) in AD and correlates well with the severity of dementia. Stages V-VI operationally define AD.

The correct answer is: Burden of neurofibrillary tangles

Question **3**

Not answered

Marked out of 1.00

Flag question

A 10 year old child in a special school has challenging behaviour associated with aggression and stereotyped motor activity. Which of the following seen in MRI is associated with autism?

Select one:

- Cerebral atrophy
- Hypoplastic cerebellum
- Medial temporal atrophy
- Caudate atrophy
- White matter hyperintensities

Hypoplasia of cerebellar vermis and to some extent the cerebellar hemispheres is documented.

The correct answer is: Hypoplastic cerebellum

Question **4**

Not answered

Marked out of 1.00

Flag question

A 33-year-old man presents with anxiety and depression. Soon he develops rapidly progressive cognitive impairment with myoclonus, ataxia and eventually akinetic mutism. Characteristic pathological change expected in brain autopsy, in this case, is

Select one:

- Lewy bodies
- Extensive vacuole formation
- Ventricular dilatation
- Lack of gliosis
- Eosinophilic extraneuronal deposits

The description suggests CJD. Microscopically CJD shows a spongiform encephalopathy secondary to neuropil vacuolisation.

The correct answer is: Extensive vacuole formation

Question **5**

Not answered

Marked out of 1.00

Flag question

Which of the following structures is larger on the left side compared to the right hemisphere in healthy right handed subjects?

Select one:

- Cingulate cortex
- Planum temporale
- Superior colliculus
- Thalamus
- Amygdala

The planum temporale shows a marked leftward volume asymmetry that is related to the degree of right-handedness; this is more marked in males (nearly ten times asymmetrical on the left in males)

The correct answer is: Planum temporale

Question **6**

Not answered

Marked out of 1.00

Flag question

Arteriosclerotic changes in major arteries are most commonly found in which of the following types of dementia?

Select one:

- Multi-infarct dementia
- Lewy body dementia
- Alzheimer's dementia
- Parkinson's disease dementia
- Normal Pressure Hydrocephalus

In multi-infarct dementia, brain damage due to multiple cerebral infarcts leads to local or general brain atrophy and ventricular enlargement. The presence of arteriosclerotic changes is commonly noted.

The correct answer is: Multi-infarct dementia

Question **7**

Not answered

Marked out of 1.00

Flag question

Alpha-synuclein is a major constituent of which of the following?

Select one:

- Neurofibrillary tangles
- Pick's bodies
- Prion particles
- Neuritic plaques
- Lewy bodies

Lewy bodies are weakly eosinophilic, spherical, cytoplasmic inclusions. Lewy bodies in Parkinson's disease and DLB contain accumulations of alpha-synuclein

The correct answer is: Lewy bodies

Question **8**

Not answered

Marked out of 1.00

Flag

Which disease can be studied by inducing 'autoimmune encephalomyelitis'?

Select one:

- Acquired brain injury
- Parkinson's disease

question

- Huntington's disease
- Multiple sclerosis
- Alzheimer's disease

Experimental autoimmune encephalomyelitis (EAE) is a laboratory model used to study the efficacy of potential agents treating multiple sclerosis (MS). EAE is also used to investigate the pathogenesis of autoimmunity, CNS inflammation and demyelination.

The correct answer is: Multiple sclerosis

Question **9**

Not answered

Marked out of 1.00

Flag question

Which of the following is correct with respect to planum temporale?

Select one:

- It is present only on dominant cerebral hemisphere
- It is normally larger on left side
- It specialises in visual processing
- In schizophrenia left More than right asymmetry is noted
- It is situated in anterior temporal surface

The planum temporale, the posterior superior surface of the superior temporal gyrus, is a highly lateralized brain structure involved with language. In schizophrenic patients, a consistent reversal of the normal left-larger-than- right asymmetry of planum temporale surface area is noted (Planum temporale asymmetry reversal in schizophrenia. Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/9137122).

The correct answer is: It is normally larger on left side

Question **10**

Not answered

Marked out of 1.00

Flag question

The neuropathological findings in Punch Drunk syndrome include which of the following changes?

Select one:

- Knife blade gyri
- Basal ganglia enlargement
- Neurofibrillary tangles
- Caudate hypertrophy
- Pulvinar signs

Dementia pugilistica (DP) is a type of neurodegenerative disease that can affect around 15% of boxers who suffer concussions after 12-16 years of boxing on average. Symptoms and signs of DP develop progressively over a long latent period.Histological changes in Punch Drunk syndrome (Dementia Pugilistica) would include neuronal loss and neurofibrillary tangles. Thinning of the corpus callosum, perforation of the septum pellucidum and

ventricular enlargement is also seen in this condition.

The correct answer is: Neurofibrillary tangles

Question **11**

Not answered

Marked out of 1.00

Flag question

Which of the following is correct concerning white matter hyperintensities seen in mood disorders?

Select one:

- Best seen in T1 images of MRI
- Seen only in bipolar patients
- Associated with good recovery from mood episodes
- Associated with vascular risk factors
- Seen more often in younger patients

A strong association between mood disorder and the number and severity of focal signal hyperintensities on T2-weighted images has been established. These white matter hyperintensities (WMH) occur particularly in the deep subcortical white matter and to a lesser extent in the basal ganglia and periventricularly. They are seen in excess in bipolar and unipolar mood disorder, with an odds ratio of 3 to 7. In major depression, WMH are particularly common in elderly subjects, where they are linked to risk factors for, and the presence of, vascular disease (Harrison, P.The neuropathology of primary mood disorder | Brain, http://brain.oxfordjournals.org/content/125/7/1428).

The correct answer is: Associated with vascular risk factors

Question **12**

Not answered

Marked out of 1.00

Flag question

Knife blade atrophy is seen in

Select one:

- Pick's disease
- Alzheimer's disease
- Lewy body dementia
- CJD
- Progressive supranuclear palsy

Knife-blade type atrophy refers to the marked atrophy of the frontal and temporal lobes with relative sparing of parietal and occipital lobes in patients with Pick's disease.

The correct answer is: Pick's disease

Question **13**

Not answered

Marked out of 1.00

Selective cerebellar atrophy as a neuropathological change is likely to be seen in which of the following dementias?

Select one:

Flag question

- Lewy body dementia
- Creutzfeldt-Jakob Disease
- Alzheimer's dementia
- Normal Pressure Hydrocephalus
- Parkinson's disease dementia

In Creutzfeldt - Jakob disease, macroscopic changes would include generalised cerebral atrophy, sometimes a selective cerebellar atrophy and ventricular enlargement. Histological changes would include status spongiosus, neuronal degeneration without inflammation and astrocytic proliferation. There is a nerve cell loss, gliosis and spongiform changes.

The correct answer is: Creutzfeldt-Jakob Disease

Question **14**

Not answered

Marked out of 1.00

Flag question

A 55-year-old lady is a known alcoholic. She drinks in binges and presents with deficits in new learning though she could retain messages long enough to hold a conversation. The most prominent pathology in this patient will be seen in

Select one:

- Hippocampus
- Dorsolateral prefrontal cortex
- Hypothalamus
- Dorsal medial thalamus
- Cerebellum

Wernicke's encephalopathy is characterized by degenerative changes including gliosis and small hemorrhages in structures surrounding the third ventricle and aqueduct (i.e. the mamillary bodies, hypothalamus, mediodorsal thalamic nucleus, colliculi, and midbrain tegmentum), as well as cerebellar atrophy.

The correct answer is: Dorsal medial thalamus

Question **15**

Not answered

Marked out of 1.00

Flag question

Marked atrophy of the caudate nucleus is seen in which of the following disorders?

Select one:

- Huntington's disease
- Punch drunk syndrome
- Alcoholic dementia
- Creutzfeldt-Jakob Disease
- Lewy Body dementia

Macroscopic findings in Huntington's disease include 1. Small brain with reduced mass 2. Marked atrophy of the corpus striatum, particularly the caudate nucleus 3. Marked atrophy of

the cerebral cortex, particularly the frontal lobe gyri 4. Dilatation of the lateral and third ventricles. Histological changes in Huntington's disease would include neuronal loss in the cerebral cortex and neuronal loss in the corpus striatum and astrocytosis in the affected regions.

The correct answer is: Huntington's disease

Question **16**

Not answered

Marked out of 1.00

Flag question

Lewy bodies are eosinophilic, haloed, intra neuronal inclusions. They contain which of the following materials?

Select one:

- Glial tissue
- Paired helical filaments
- Dopamine metabolites
- synnuclein
- Amyloid substance

Lewy bodies contain synnuclein, Ubiquitin, protein neurofilaments, granular material, dense core vesicles and microtubule assembly protein. In Lewy body dementia, the density of Lewy bodies is much higher in the cingulate gyrus, parahippocampal gyrus and temporal cortex.

The correct answer is: synnuclein

Question **17**

Not answered

Marked out of 1.00

Flag question

Which of the following is a feature more likely to be seen in Pick's disease than other dementias?

Select one:

- Reactive astrocytosis
- Gliosis
- Tau proteins
- Ventricular shrinkage
- Knife blade gyri

Pick's disease is associated with selective asymmetrical atrophy of the frontal and anterior temporal lobes. Patients also show knife blade gyri and ventricular enlargement. The histological changes in Picks disease would include Pick's bodies, neuronal loss and reactive astrocytosis, and these changes may be seen in the cerebral cortex, substantia nigra, locus coereleus and basal ganglia. Pick bodies are round argyrophilic intraneuronal inclusions and pick cells are swollen cortical pyramidal cells.

The correct answer is: Knife blade gyri

Question **18**

Which of the following is a true statement concerning human HIV disease?

Not answered

Marked out of 1.00

Flag question

Select one:

- Psychosis is the most common HIV related psychiatric problem.
- CSF viral load is the best predictor of HIV related CNS disease.
- HIV cross the blood brain barrier using macrophages
- CD4 count is the better indicator for severity than viral load
- HIV does not cause programmed cell death of neurons

In order to enter the brain, HIV-1 must cross the BBB using mechanisms that remain unclear. The generally accepted model is the 'Trojan Horse hypothesis'. HIV enters the CNS as a passenger in cells trafficking to the brain via CD4 T cells or monocytes (Retrovirology: HIV-1 associated dementia. Retrieved from http://www.retrovirology.com/content/3/1/28).

The correct answer is: HIV cross the blood brain barrier using macrophages

Question **19**

Not answered

Marked out of 1.00

Flag question

The enzyme involved in the metabolism of amyloid precursor protein that prevents amyloid formation is

Select one:

- Protein kinase
- Delta-secretase
- Gamma-secretase
- Alpha-secretase
- Beta-secretase

Alpha secretases are a family of proteolytic enzymes that cleave amyloid precursor protein in its transmembrane region. In conditions like Alzheimer's disease, when amyloid precursor protein is processed by beat-secretase and gamma secretase, it gives rise to beta amyloid peptide, which plays a crucial role in the pathogenesis of Alzheimer's dementia.

The correct answer is: Alpha-secretase

Question **20**

Not answered

Marked out of 1.00

Flag question

In those with Mild Cognitive Impairment, which of the following biomarkers can detect worsening cognitive function?

Select one:

- Decreased tau protein in CSF
- Increased Tau-to-Amyloid ratio in CSF
- Decreased Ubiquitin-to-Amyloid ratio in CSF
- Decreased Tau-to-Amyloid ratio in CSF
- Increased beta amyloid in CSF

Individuals with early Alzheimer's dementia have reduced CSF beta-amyloid and increased levels of CSF tau and phosphorylated tau. The CSF tau/amyloid beta ratio is nearly 5 times higher in those who later convert to full-blown Alzheimer's dementia compared to non-converters. Increased CSF tau/A-beta (amyloid) ratios show strong promise as a preclinical biomarker to predict future dementia in cognitively normal older adults. Fagan et al., Arch Neurol. 2007 Mar;64(3):343-9.

The correct answer is: Increased Tau-to-Amyloid ratio in CSF

Question **21**

Not answered

Marked out of 1.00

Flag question

A 40-year-old woman was found wandering on streets after the recent funeral of her husband. She can give her personal demographic details and has no past psychiatric history. She has no physical injuries. Which of the following deficits is likely? (June 2008)

Select one:

- Loss of semantic memory
- Loss of new learning capacity
- Continuous anterograde amnesia
- Loss of procedural memory
- Loss of memory of personal events well rehearsed

This is characteristic of dissociative amnesiWell rehearsed personal events are lost (such as the fact that her husband was ill before death, for example).

The correct answer is: Loss of memory of personal events well rehearsed

Question **22**

Not answered

Marked out of 1.00

Flag question

You have noticed some degree of cognitive decline and weight loss in an intravenous drug user. The most probable diagnosis is

Select one:

- Neurosyphilis
- Alzheimer's dementia
- CADASIL
- Binswanger's disease
- HIV dementia

One of the common sources of acquiring HIV infection is intravenous drug use. Subcortical dementia is common in patients with HIV in late stages of their illness

The correct answer is: HIV dementia

Question **23**

Not answered

Marked out of

Which of the following best describes the difference between CJD and vCJD? The variant CJD has

1.00

Flag question

Select one:

- Earlier onset of ataxia
- Negative tonsillar biopsy
- Absence of florid plaques on autopsy
- Earlier age of onset
- Typical EEG changes

In 1996, a new type of CJD called vCJD (variant CJD) was reported. Variant CJD appears to affect younger people than the other forms of the disease, with an average age of death of 29 years. Death occurs on an average within 14 months.

The correct answer is: Earlier age of onset

Question **24**

Not answered

Marked out of 1.00

Flag question

Hirano bodies are seen in brain autopsy of a patient who had a history of cognitive impairment. Which of the following diagnosis is most likely?

Select one:

- Vascular dementia
- Wilson disease
- Crutzfeld Jakob disease
- Alzheimer's dementia
- Lewy body dementia

Hirano bodies are rod-shaped eosinophilic bodies in the cytoplasm of neurons that may be set free in the extracellular space if the neuron dies. Hirano bodies seen in Alzheimer's are intracellular aggregates of actin and actin-associated proteins. They are frequently seen in hippocampal pyramidal cells

The correct answer is: Alzheimer's dementia

Question **25**

Not answered

Marked out of 1.00

Flag question

Which one of the following is a protective factor against Alzheimer's disease?

Select one:

- Down's syndrome
- Apolipoprotein e2 allele
- Head injury
- Post menopausal estrogen decline
- Age

Apolipoprotein e2 allele is a proven protective factor against Alzheimer's disease. Other possible protective factors are smoking, NSAIDs, oestrogen, premorbid intelligence and

education.

The correct answer is: Apolipoprotein e2 allele

Question **26**

Not answered

Marked out of 1.00

Flag question

Depigmentation of the substantia nigra is seen in which of the following disorders?

Select one:

- Creutzfeldt-Jakob Disease
- Huntington's disease
- Dandy-Walker malformation
- Alzheimers disease
- Parkinson's disease

The macroscopic changes in idiopathic Parkinsons disease would include depigmentation of the substantia nigra, particularly the zona compacta, and depigmentation of the locus coereleus and diffuse cortical atrophy. The histological changes in idiopathic Parkinsons disease would include Reactive astrocytosis, neuronal loss, Presence of Lewy bodies in different areas such as the substantia nigra, locus cereleus, dorsal motor nucleus of the vagus, hypothalamus, nucleus basalis of meynert, Edinger-Westphal nucleus and raphe nuclei.

The correct answer is: Parkinson's disease

Question **27**

Not answered

Marked out of 1.00

Flag question

The spongiform appearance of brain tissue in CJD is due to

Select one:

- Apoptosis of neuronal cells
- Glial proliferation
- Demyelination
- Neuropil vacuolation
- Excessive fibrosis

Microscopically CJD shows a spongiform encephalopathy secondary to neuropil vacuolisation.

The correct answer is: Neuropil vacuolation

Question **28**

Not answered

Marked out of 1.00

Flag question

In which of the following conditions both balloon cells and Hirano bodies are seen?

Select one:

- Alzheimer's disease
- HSE encephalitis

- Neurosyphilis
- CJD
- Pick's disease

Hirano bodies are fusiform or spheroidal eosinophilic bodies, that are commonly observed in the hippocampi of the elderly and are especially numerous in patients with various dementias or degenerative diseases (Ultrastructure of Hirano bodies - Springer. Retrieved from http://link.springer.com/article/10.1007%2FBF00687999). Both Hirano bodies and Balloon cells are seen together in Pick's disease.

The correct answer is: Pick's disease

Question **29**

Not answered

Marked out of 1.00

Flag question

The immunological staining used in detection of Pick's disease is

Select one:

- Glycosaminoglycan.
- Amyloid P
- Basic fibroblast growth factor antibodies
- Anti-tau antibodies
- Heparan sulfate

A variety of stains can be used to visualise Pick bodies and Pick cells, but the most efficient and highly specific method is to use immunohistochemical staining with anti-tau and anti-ubiquitin antibodies. The neurofibrillary tangles of Alzheimer's can be stained with antibodies to basic fibroblast growth factor, amyloid P, and heparan sulfate glycosaminoglycan.

The correct answer is: Anti-tau antibodies

Question **30**

Not answered

Marked out of 1.00

Flag question

Autopsy of an elderly man who suffered from marked tremors and a progressive cognitive decline shows cells negative for tau protein. What is the likely diagnosis?

Select one:

- Lewy body dementia
- Vascular dementia
- Alzheimer's dementia
- CJD
- Frontal lobe degeneration

Lewy bodies that are seen in diffuse Lewy body disease (dementia), rarely have any tau protein deposits. They are primarily made of alpha synuclein and ubiquitin which are tau negative.

The correct answer is: Lewy body dementia

Question **31**

Not answered

Marked out of 1.00

Flag question

The macroscopic changes seen in Alzheimer's disease includes

Select one:

- Senile plaques
- Neuronal loss
- Ventricular enlargement
- Vacuolisation
- Neurofibrillary tangles

The macroscopic changes in Alzheimers disease would include Enlargement of the lateral and third ventricles, global brain atrophy, reduction in brain weight and sulcal widening. Neuronal loss, senile plaques, neurofibrillary tangles, reactive astrocytosis and shrinking of dendritic branching are the microscopic changes seen in the cerebral cortex in Alzheimers disease. Cerebral atrophy and ventricular enlargement are macroscopic changes.

The correct answer is: Ventricular enlargement

Question **32**

Not answered

Marked out of 1.00

Flag question

Neurofibrillary tangles are most commonly present in which of the following locations in Alzheimer's?

Select one:

- Brain stem
- Auditory association cortex
- Hippocampus
- Prefrontal cortex
- Insula

Neurofibrillary tangles occur with ageing and in all the other conditions listed above. NFTs are found in the hippocampus (Ammons horn) and are also found in the Amygdala, parahippocampal gyrus, neocortex, locus coereleus, and nucleus of meynert and Raphe nuclei. Senile Plaques are found in the same sites as those of NFTs. Plaques include extra cellular Amyloid as a component, and beta A-4 protein is a major biochemical component of plaque Amyloid.

The correct answer is: Hippocampus

Finish review

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